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Celestia: Language Tutorial

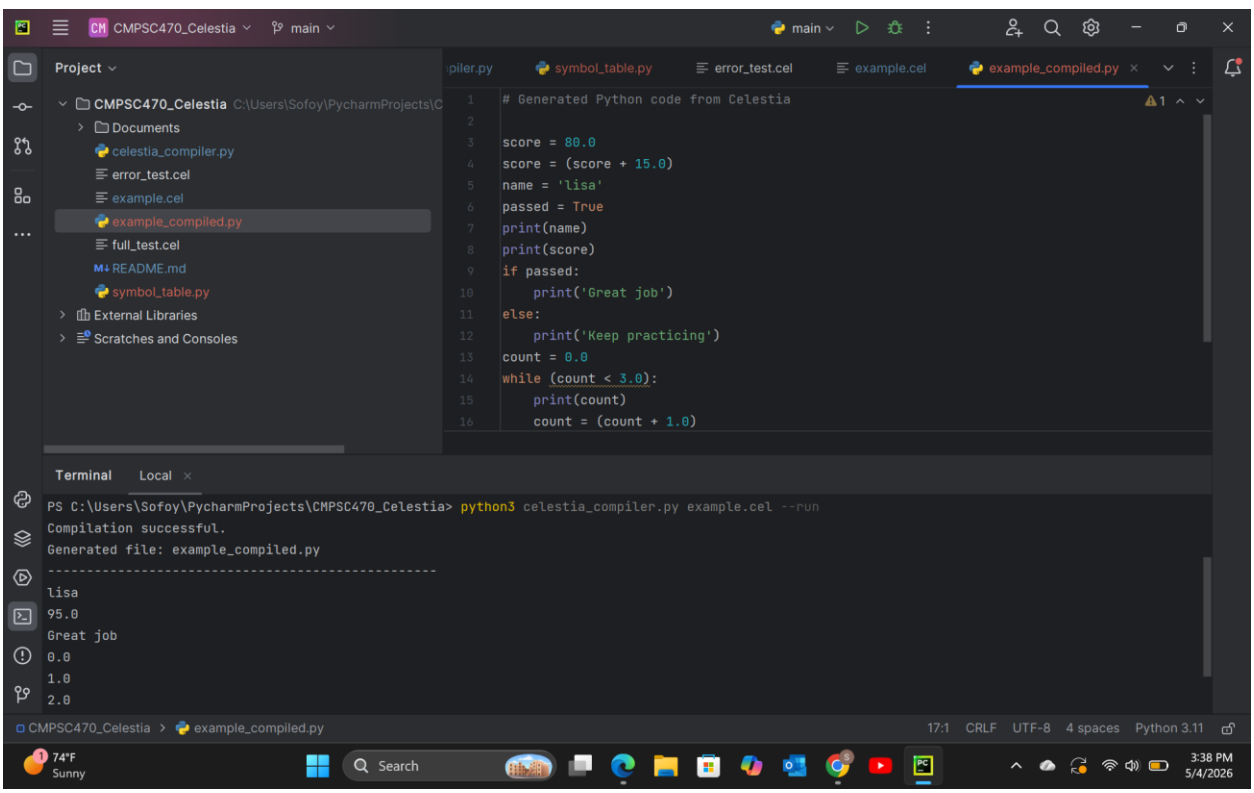
1. Overview

Celestia is a beginner-friendly toy programming language. This tutorial walks through the core features of the language with examples. By the end, you will be able to write a complete Celestia program that declares variables, performs arithmetic, uses conditionals, and loops.

2. Running a Celestia Program

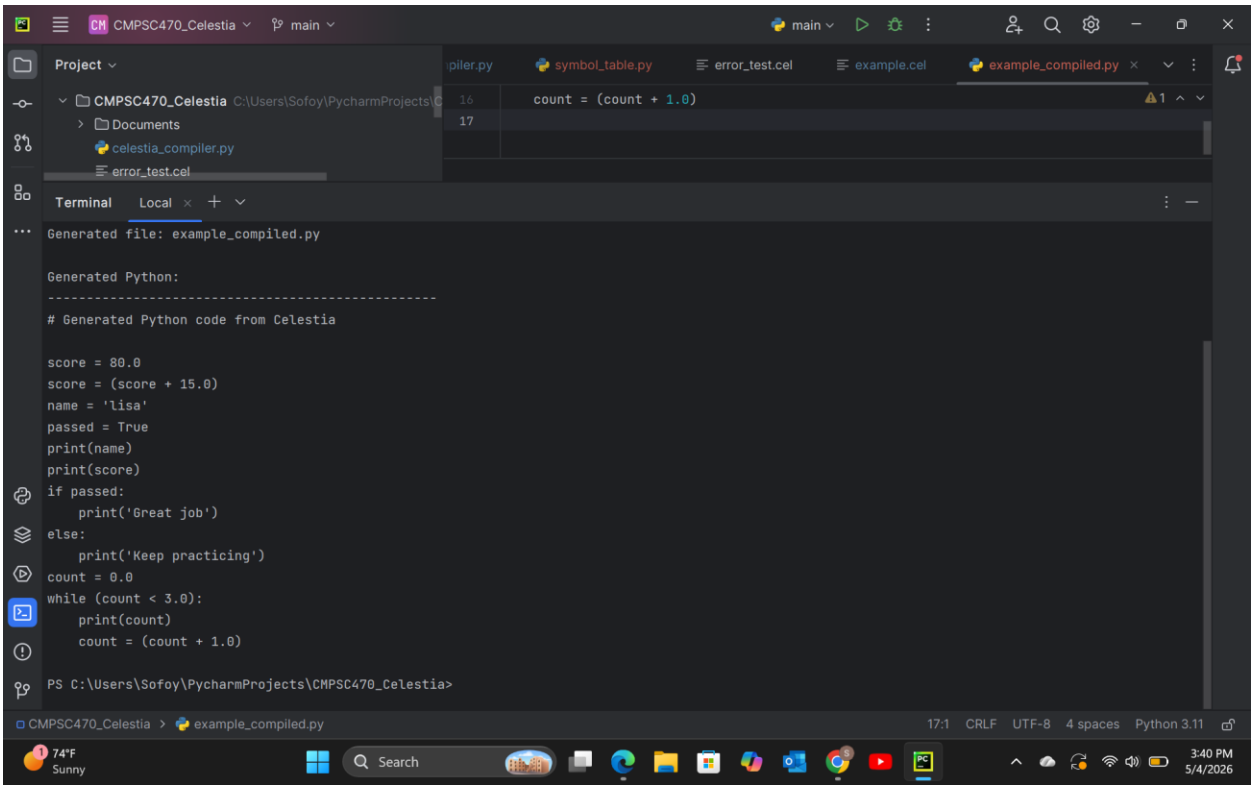
Save your code in a file with the .cel extension, then run it from the terminal:

```
python3 celestia_compiler.py example.cel --run
```



This compiles your Celestia code into Python and executes it. You can also view the generated Python code:

```
python3 celestia_compiler.py example.cel --show
```



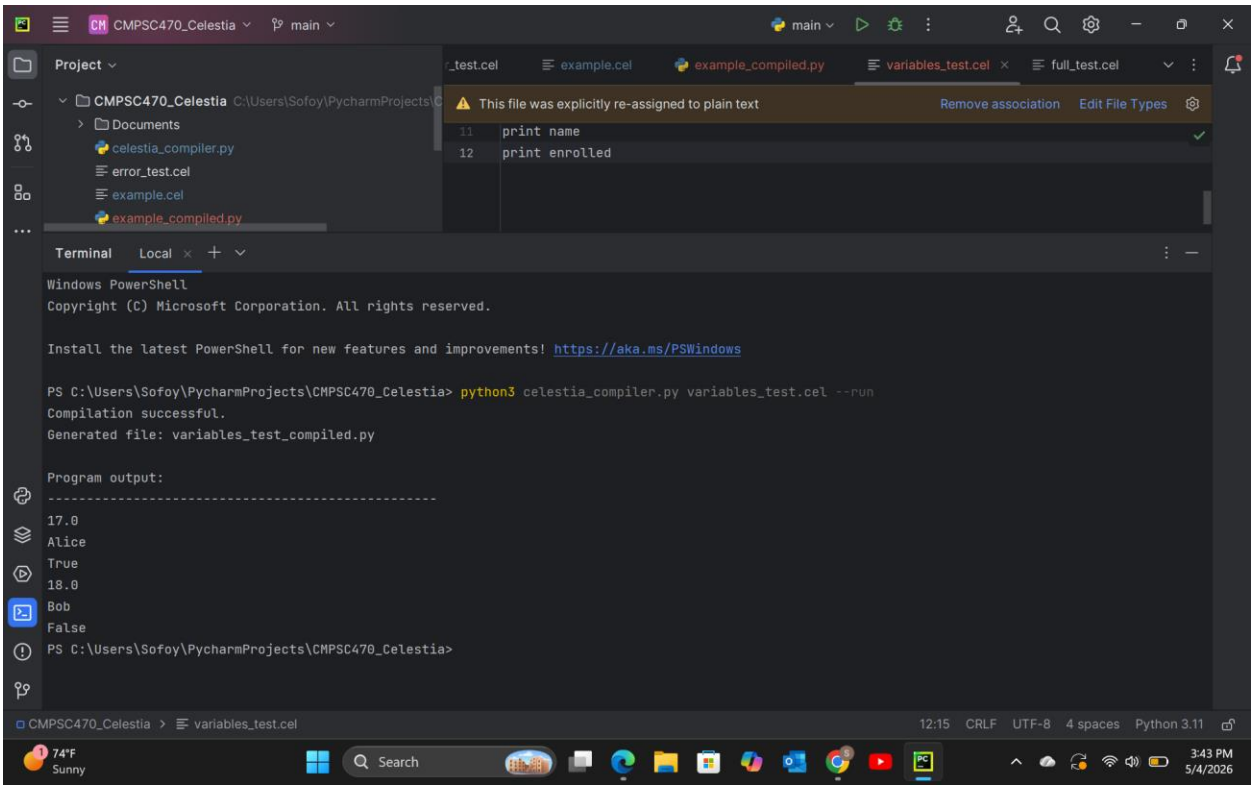
3. Declaring Variables

Celestia has three data types: num, String, and Boolean. Every variable must be declared with its type before use.

```
num age = 17
String name = "Alice"
Boolean enrolled = true
```

Once declared, a variable can be updated by assigning a new value of the same type:

```
age = 18
name = "Bob"
enrolled = false
```



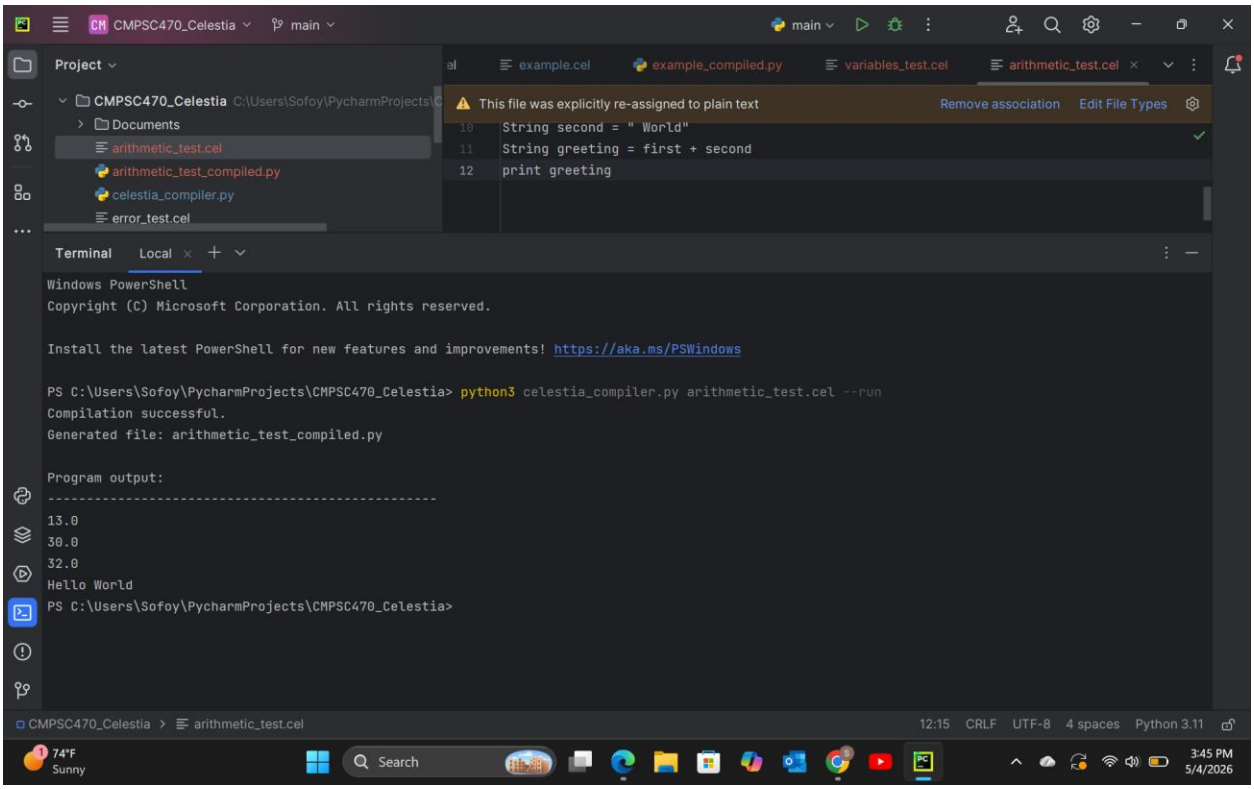
4. Arithmetic Expressions

Celestia supports +, -, *, / on num variables. You can use variables and literals together in expressions.

```
num x = 10
num y = 3
num total = x + y
num product = x * y
num result = x * y + 2
```

String variables can be concatenated using the + operator:

```
String first = "Hello"
String second = " World"
String greeting = first + second
```

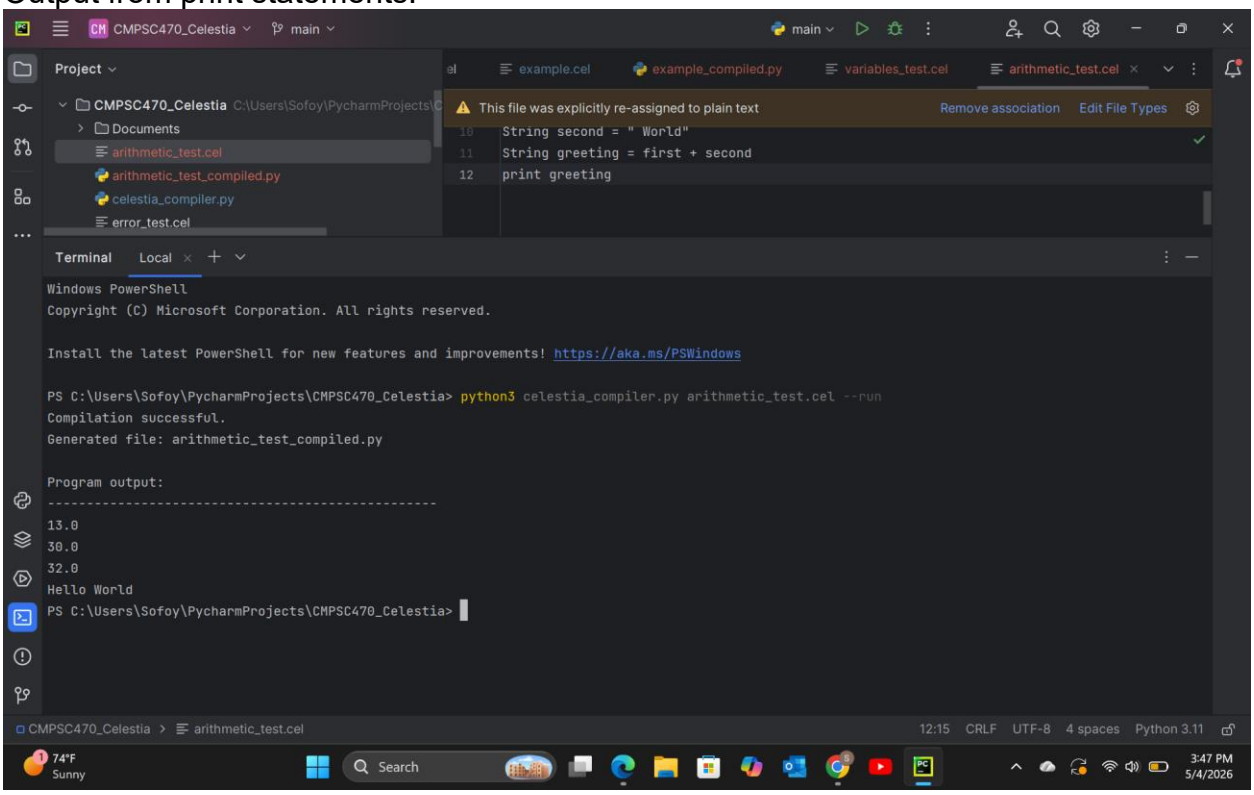


5. Printing Output

Use the print statement to display a value:

```
print age
print greeting
print "Score: "
print x + 5
```

Output from print statements:



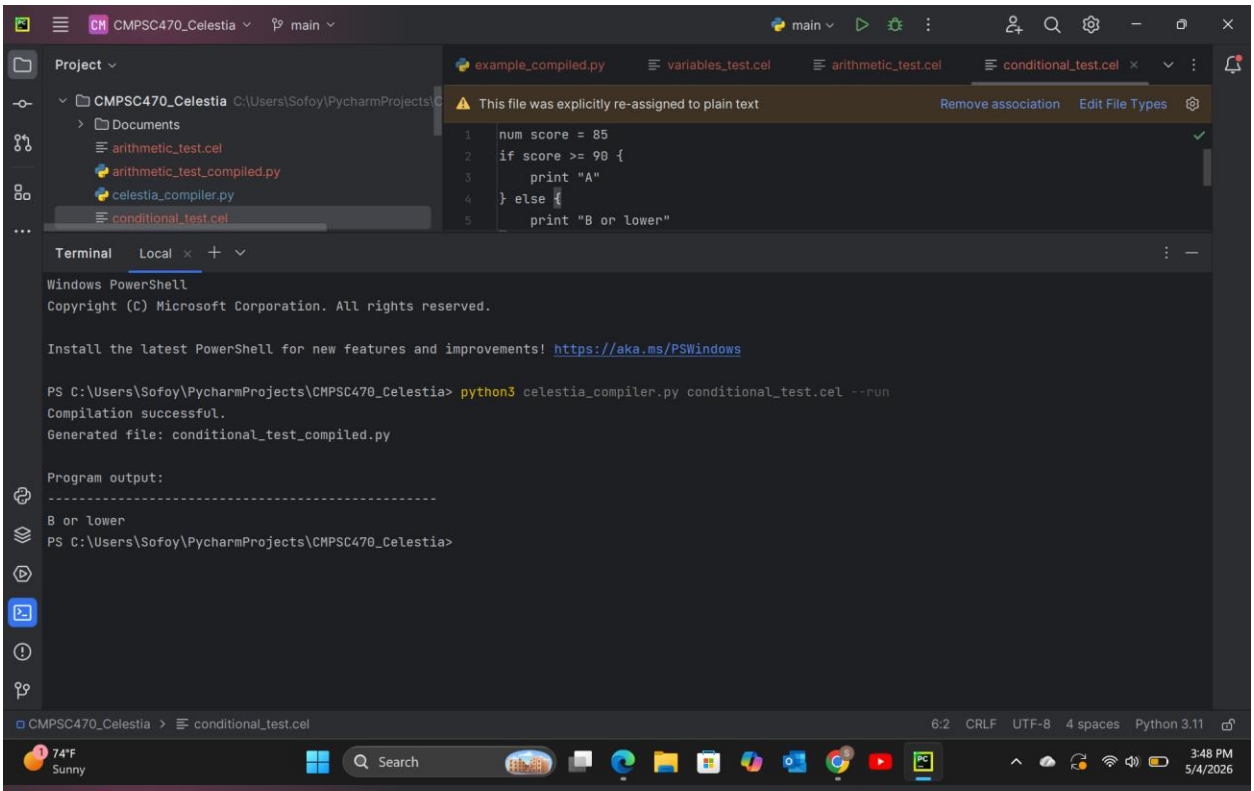
6. Conditional Statements

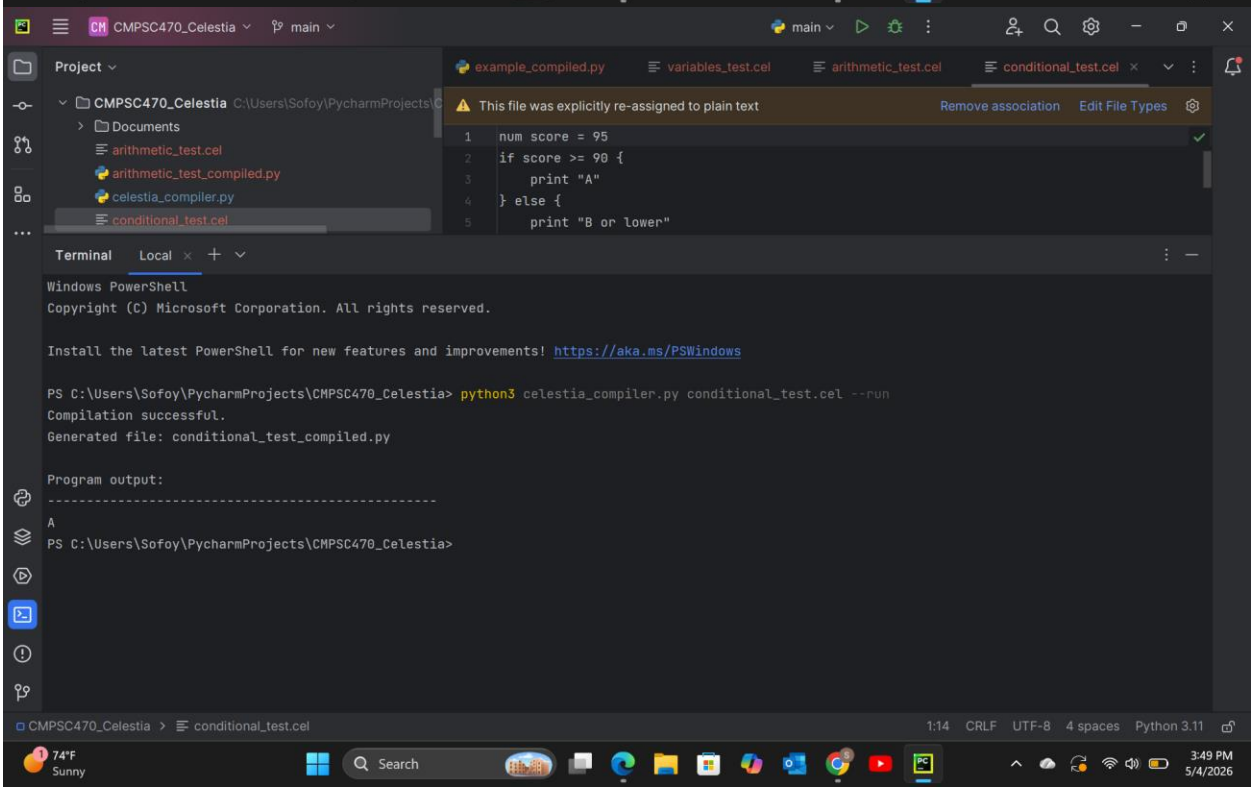
The if statement evaluates a Boolean condition. If true, the block inside the braces executes. The else block is optional.

```
num score = 85

if score >= 90 {
    print "A"
} else {
    print "B or lower"
}
```

Supported comparison operators: ==, !=, <, >, <=, >=.



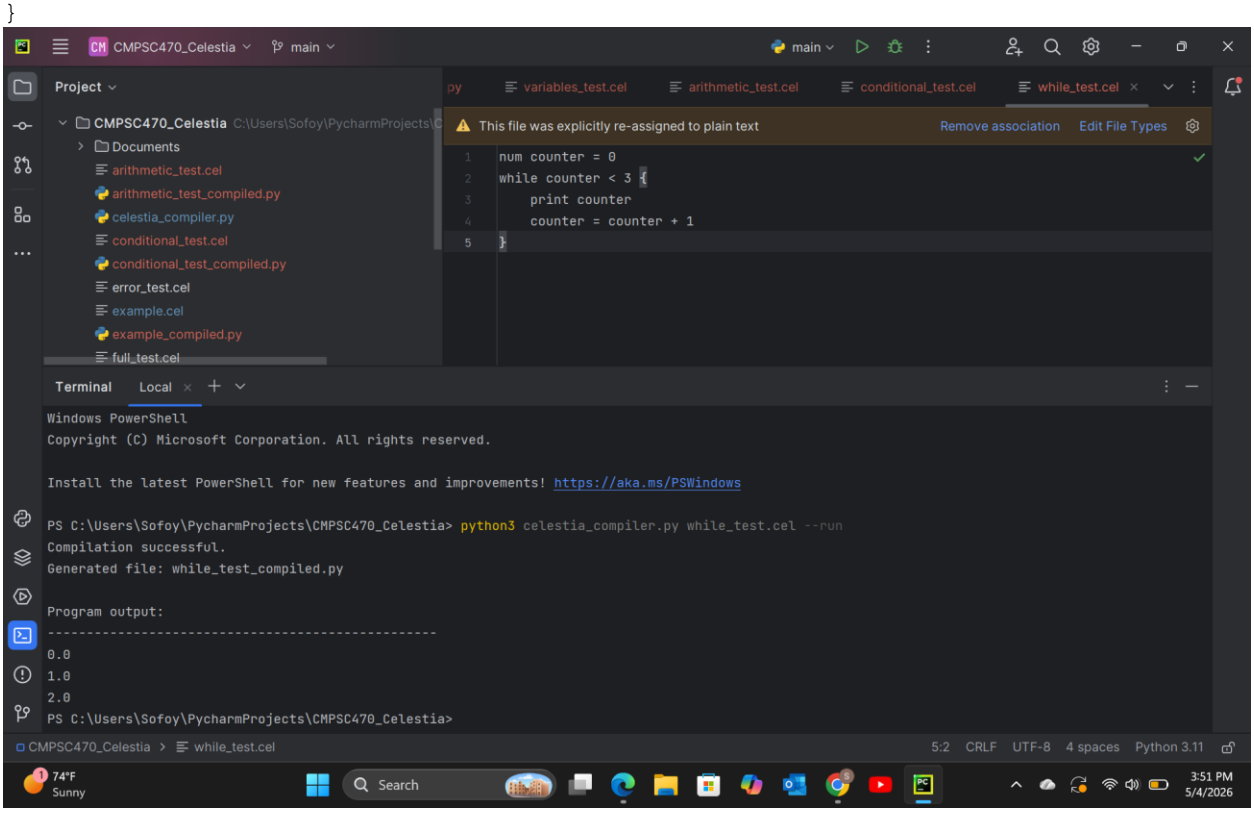


7. While Loops

The while loop repeats its body as long as the condition is true.

```
num counter = 0

while counter < 3 {
    print counter
    counter = counter + 1
}
```

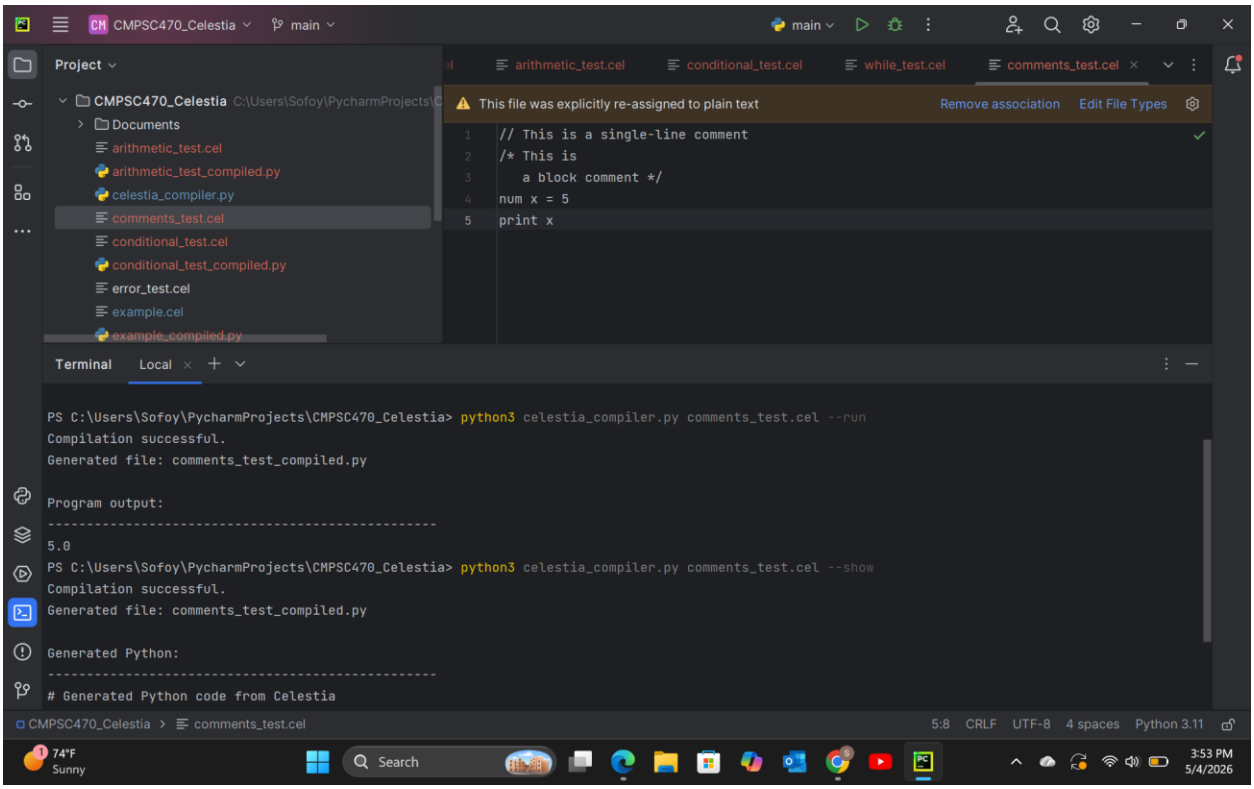


8. Comments

Single-line comments start with //. Block comments use /* and */.

```
// This is a single-line comment

/* This is
   a block comment */
```



9. Example Exercise

Write a Celestia program that does the following:

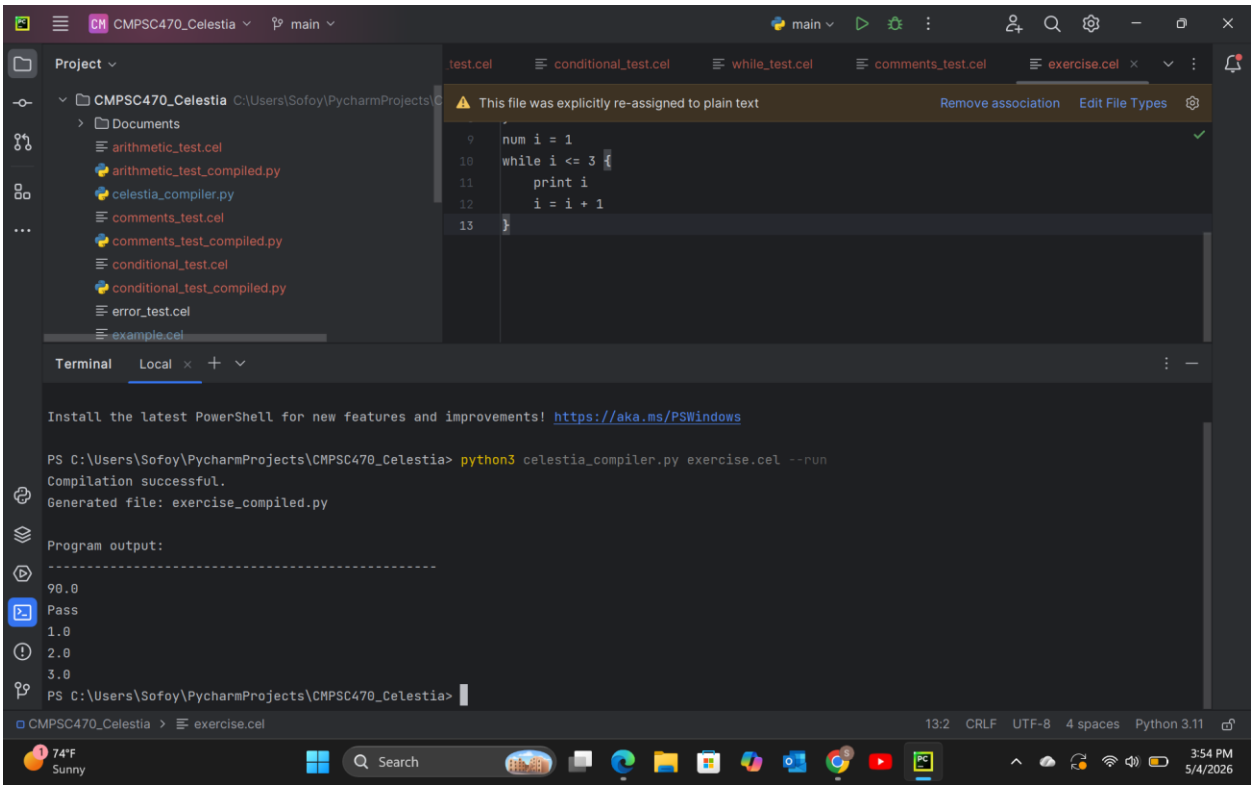
1. Declare a num variable called score and set it to 70.
2. Add 20 to score.
3. Print score.
4. If score is 90 or higher, print "Pass". Otherwise, print "Fail".
5. Use a while loop to print the numbers 1, 2, and 3.

Solution

```
num score = 70
score = score + 20
print score

if score >= 90 {
    print "Pass"
} else {
    print "Fail"
}

num i = 1
while i <= 3 {
    print i
    i = i + 1
```

}

Expected Output